Massachusetts Institute of Technology

DarkLight Simulation/Analysis Meeting
Update on Møller Scattering

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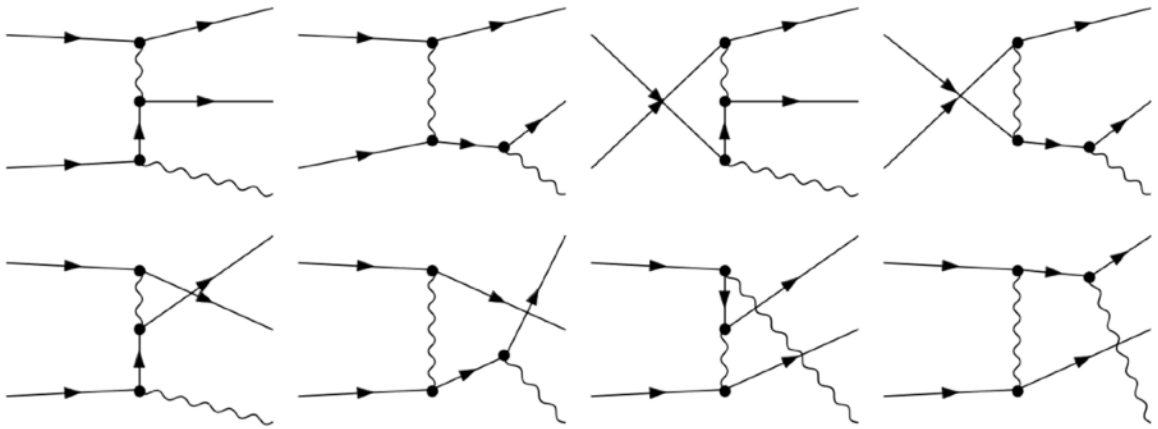
March 30, 2026

Møller Scattering

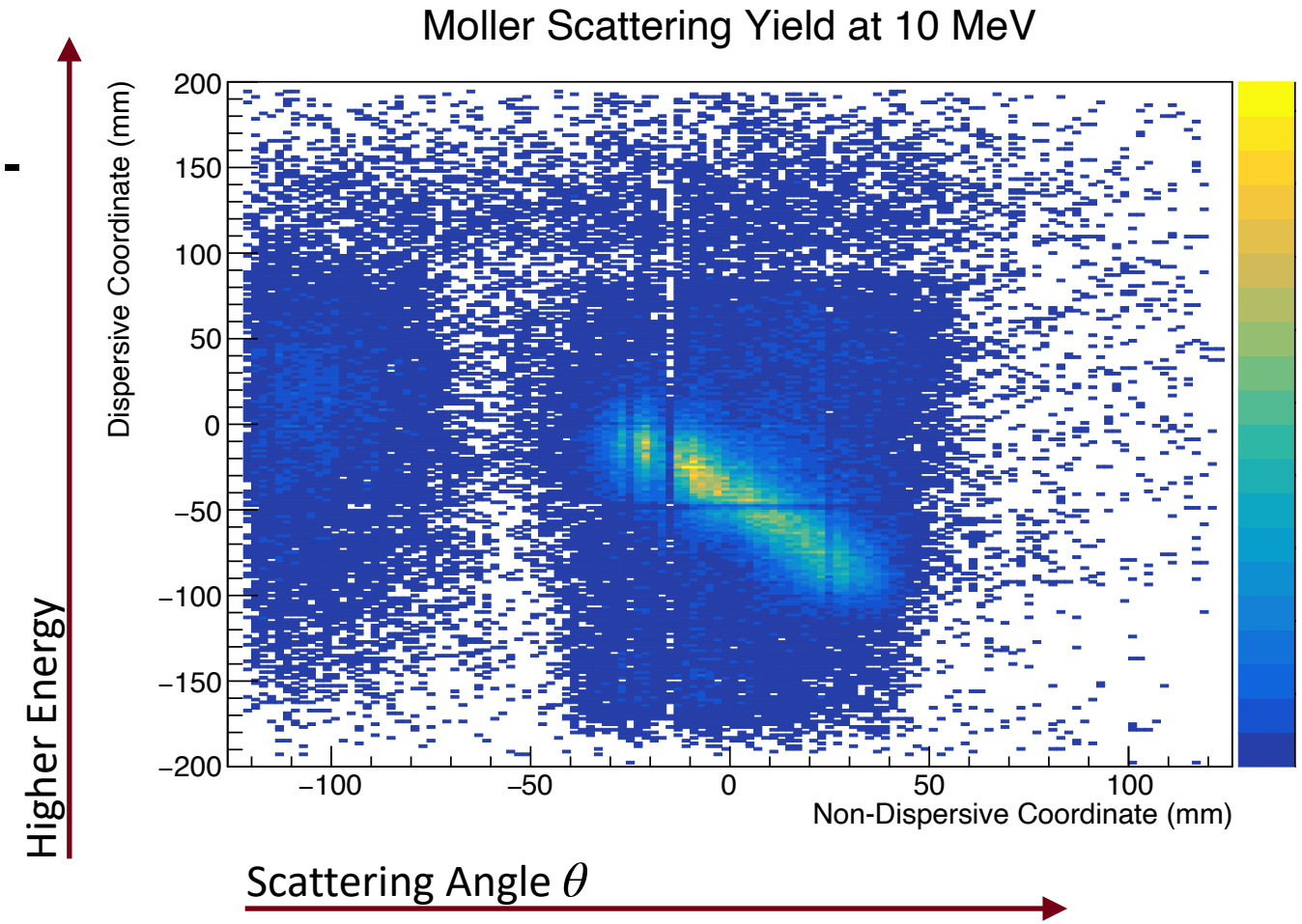
- DarkLight performed measured of radiative Møller scattering during commissioning beam time
- Data taken at 10, 15, 20, 25, and 30 MeV
- Have event generator on hand from C. Epstein and R. Milner
- Other generators exist, can test as well
- Testing in a regime where electron mass is not negligible



Radiative Møller diagrams. From https://journals.aps.org/prd/abstract/10.1103/PhysRevD.94.033004

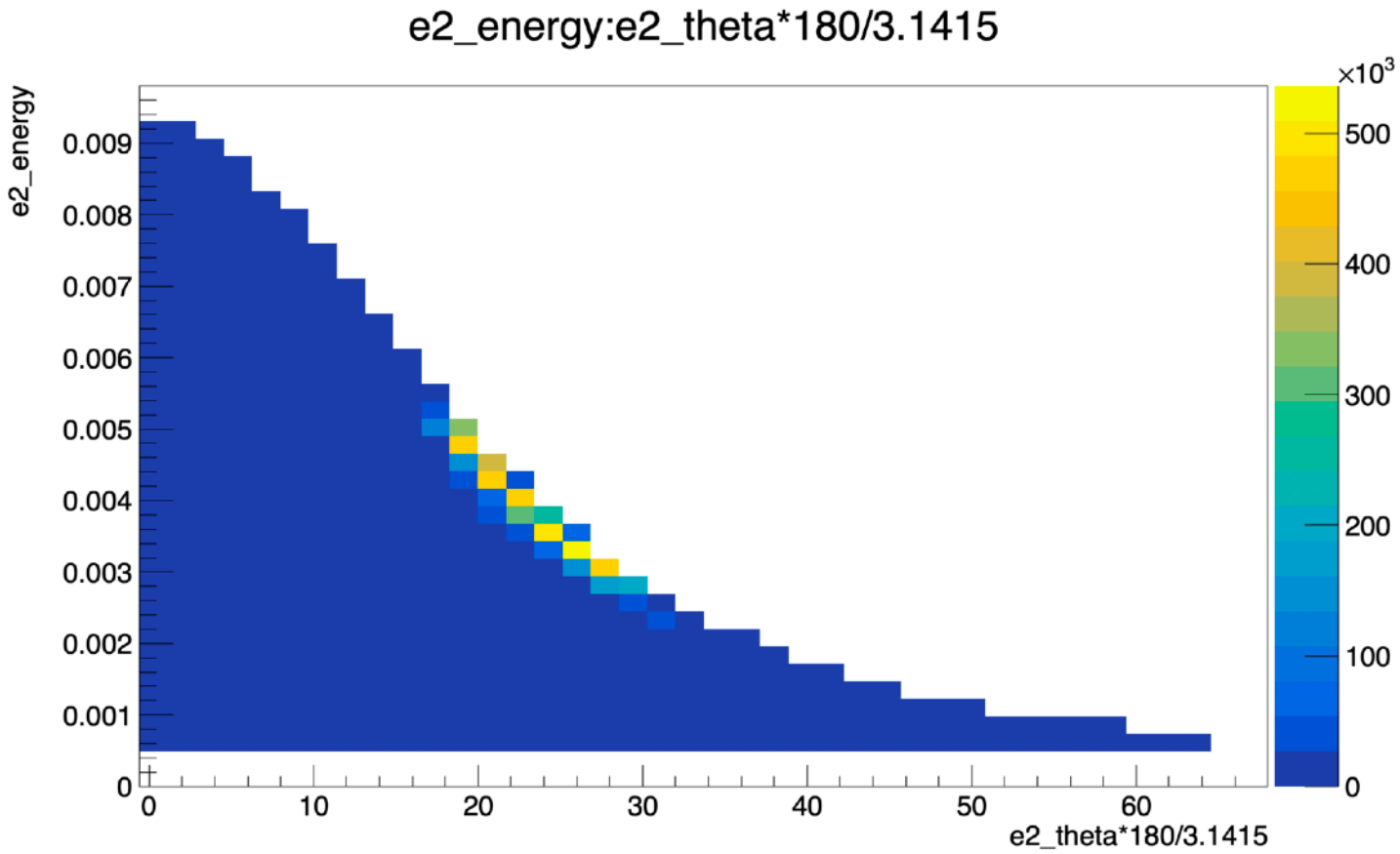
Data for 10 MeV Møller Scattering

- **Showing yield from 10 MeV Møller scattering - Right Bottom GEM**
- **Same trend for all energies, background varies depending on beam setting**
- **Would like to compare to event generator**
- **Lots of detector artifacts, dead/inefficient strips in GEMs, questions about clustering, etc.**



FIRST LOOK AT SIMULATION
Nominally everything is in a "default" setting

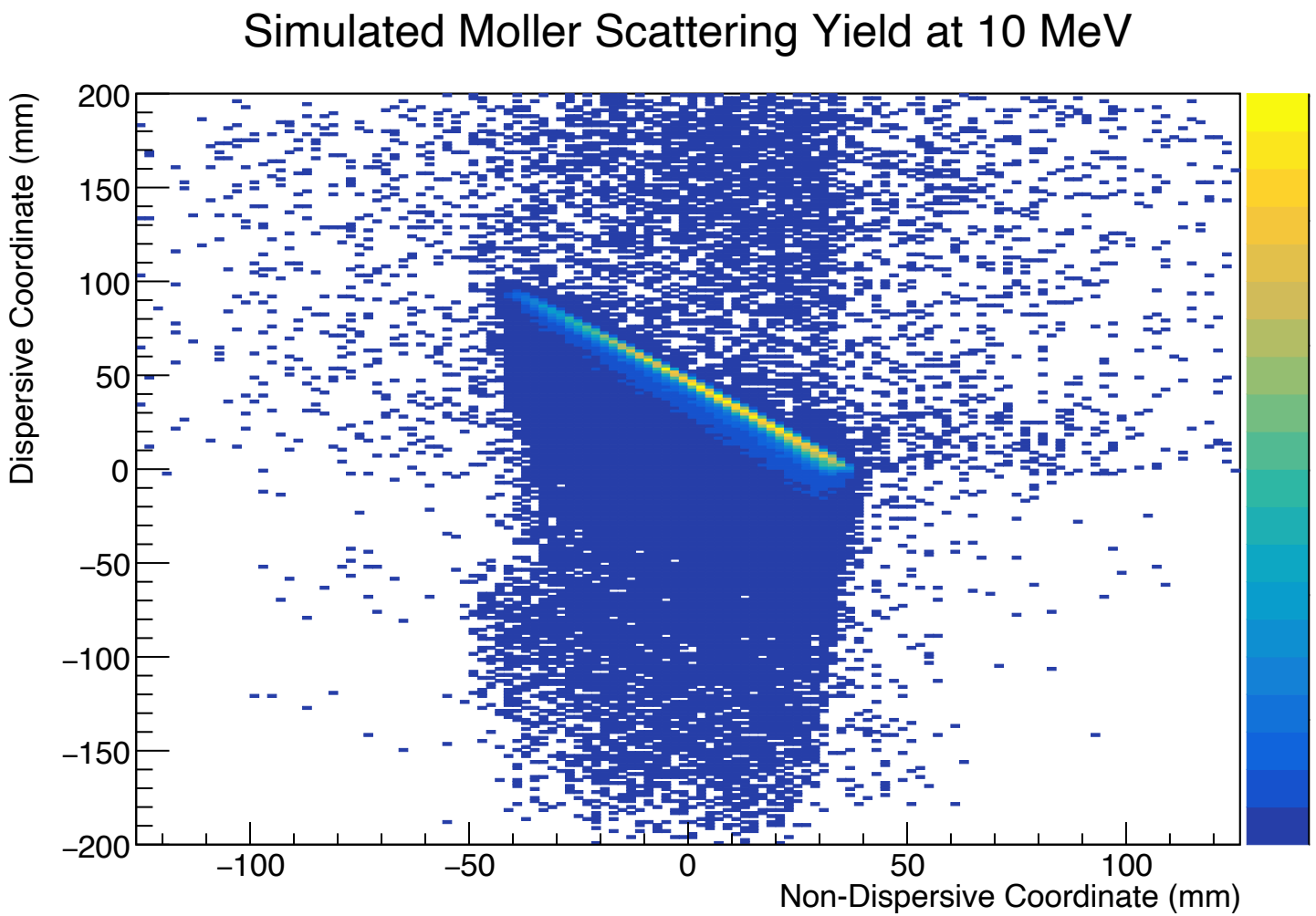
- Run generator in ~beam settings
  - 10 MeV
- Leave other settings in default configuration
  - Luminosity 1e36
  - Radiative fraction 0.75
  - dE_frac 1e-4
- Apply angular cuts in CoM frame (rads) to maximize stats going into detector
  - $1.0 < \theta < 1.5$, $-0.1 < \phi < 0.1$



Only accept into right side spectrometer - Geant4 simulation quirk

Analyze Simulation

- **Put generator output through digitization, and then standard analysis**
- **Looking at tracks in GEMs**
- **Field in simulation and in data not matched**
- **Assumed a central magnet momentum of 4.5 MeV, likely correct within ~30 %**
- **Slope in sim and data in same direction!**



Next Steps

- Implement observed inefficient/dead GEM strips
- Look into trigger analysis
  - Correlate trigger-GEM hits
- BUG BOUNTY
  - Running cooker multi-core on my Mac shows the standard setup output for multi-core, but eats the progress info. Still runs but no ETA displayed.
  - REWARD: TBD?

```
--Define Histograms--
--Startup--
Heads Up: didn't find cuts directory
R2 SLOWCTRL: Could not find index map
   setCutD "gemini:example",-100................done
Configuring slowctrl ...
................done
--Define Histograms--
--Startup--
Heads Up: didn't find cuts directory
R1 SLOWCTRL: Could not find index map
--Looping--
--Looping--
--Looping--
--Looping--
--Looping--
```